

Course Code : ECT 453-2

GVHW/RW – 23 / 1031

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

LONG TERM EVOLUTION TECHNOLOGIES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Solve all questions.
- (2) Draw the diagrams wherever necessary.

1. (a) Discuss 3GPP evolution toward an IP based flat LTE SAE architecture. 5(CO1)
(b) Give an account of multiple antenna techniques used in LTE. 5(CO2,3)
2. (a) What is the minimum and maximum data rate loss due to the cyclic prefix ?
What is the minimum theoretical loss if a shorter cyclic prefix could be used ? 5(CO3,4)
(b) Discuss the operation of OFDM transmitter and Receiver. 5(CO3,4)
3. (a) Draw the structure of the downlink resource grid and explain in brief. 5(CO2,3)
(b) Discuss the problem of PAPR in LTE downlink. 5(CO2,3)
4. (a) Illustrate the radio interface protocol architecture and the SAPs between different layers of LTE. 5(CO4)
(b) Statement : LTE supports 10 ms frames and 1 ms sub-frames.
Describe the above statement with suitable diagram(s) and appropriate justification. 5(CO1,3)

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Contd.

5. (a) Illustrate with suitable diagram the mobility management over S1 Interface. 5(CO4)
- (b) To efficiently support the varying QoS requirements of different IP applications, LTE uses the concept of a bearer as the central element of QoS control. Elaborate the statement with suitable diagram and explanation. 5(CO3,4)
6. How LTE-A technology is different than LTE and its predecessors ? 10(CO5)

